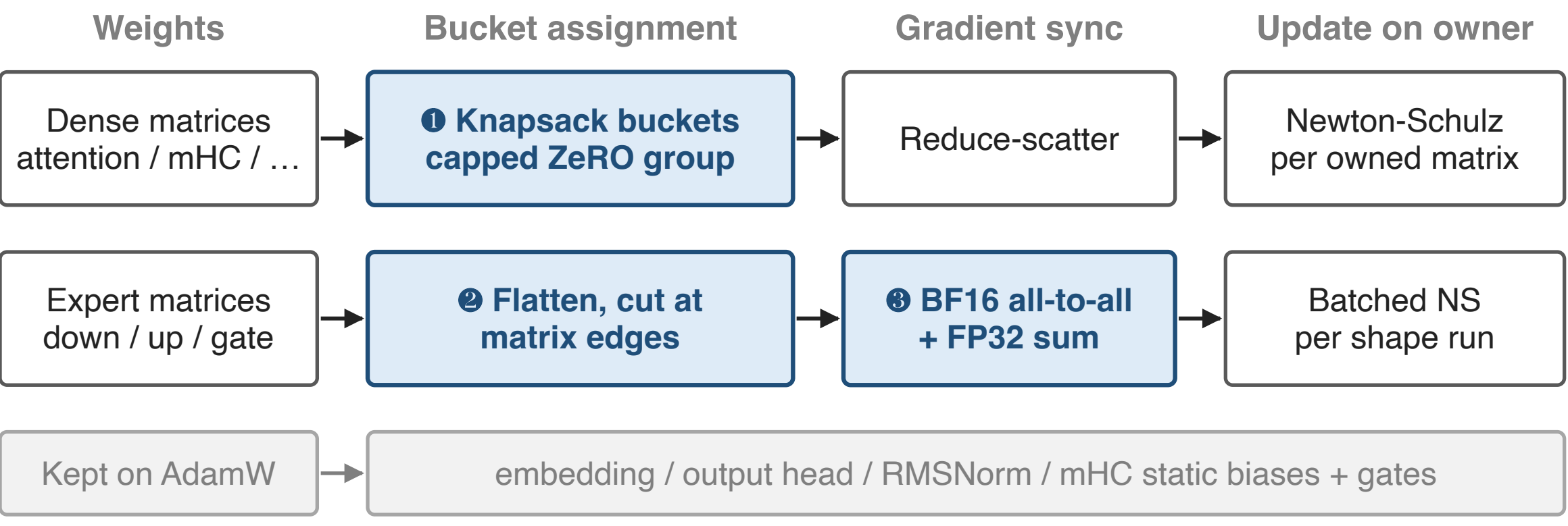
Weights
Bucket assignment
Gradient sync
Update on owner
Dense matrices attention / mHC / …
❶ Knapsack buckets capped ZeRO group
Reduce-scatter
Newton-Schulz per owned matrix
Expert matrices down / up / gate
❷ Flatten, cut at matrix edges
❸ BF16 all-to-all + FP32 sum
Batched NS per shape run
Kept on AdamW
embedding / output head / RMSNorm / mHC static biases + gates